13.7.2 Stabilization with Asphalt Materials

Asphalt works best where minimal plastic clay fines are present. If sampling and testing an existing pavement reveals that some plastic clay is present, it is may be possible to treat the material with lime to reduce the plasticity in preparation for asphalt stabilization. Stabilization with asphalt materials should not be used when the PI of the soils being stabilized exceeds 10 or the minus 200 (0.075 millimeter) material exceeds 25 percent.

Emulsified Asphalt

Various asphalt emulsion binders have been used for FDR. Mixing grade asphalt emulsions (slow set cationic or anionic) are usually used, with the specific type dependent on the characteristics of the materials on site. To increase the initial strength and resistance to water damage, small amounts (1–1.5 percent) of either lime or cement can be added to a mix using an asphalt emulsion. They can be added as dry material or as slurry (lime, cement, and water). The mix design procedures discussed for use with cold-recycling should be followed when asphalt emulsions are used for FDR.

Foamed Asphalt

Foamed asphalt, also used for FDR, is created when small amounts of water and compressed air come into contact with a hot asphalt binder in a specially designed foaming nozzle. As the water vaporizes, it expands to create tiny bubbles within the asphalt binder, decreasing the asphalt binder's viscosity and providing an appearance similar to black shaving cream. Only 1–3 percent water by weight of the asphalt is required to foam an unmodified PG asphalt binder. The tiny foam bubbles carry a thin film of asphalt that interacts with fine aggregates to create a mortar that coats large particles. The water flashes to steam immediately during the reaction and is not part of the asphalt binder.

Foam-stabilized material can be placed, shaped, compacted, and opened to traffic shortly after mixing. It remains workable for a period of time and can be stockpiled and placed at a later date by including additional moisture for compaction. Generally, the material being treated with foamed asphalt should have between 5 and 20 percent minus 200 materials. Fines that exist in most crushed-limestone base materials or that are created in the reclaiming process from soil-cement base are ideal for use with foamed asphalt.

Mix Design with Foamed Asphalt

Foamed asphalt is characterized by *expansion ratio* and *half-life*. Expansion ratio is defined as the ratio between the maximum volume achieved by the foamed asphalt binder and the volume of the asphalt binder after the foam has completely subsided. Half-life is the time required, in seconds,

for the foam to settle to half the maximum volume attained. The foaming characteristics of the binder are influenced by these factors:

- Temperature of the asphalt binder—typically about 320°F (160°C);
- Amount of water added to the asphalt binder—typically 2 percent plus or minus 1 percent by weight of the asphalt binder;
- Pressure used to inject the asphalt binder into the expansion chamber—should be about 45 psi (3 bars); and,
- Presence of antifoaming agents, such as silicone additives, in the asphalt binder. These materials should not be added to an asphalt binder that will be used as a foamed asphalt.

The selection of the asphalt binder for a project is based on the in-service environmental characteristics of the pavement being built as recommended in Chapter 2 "Petroleum Asphalt."

The first step in the mix design process is to optimize the foaming characteristics of the foamed asphalt. The Iowa DOT has developed an excellent procedure for designing foamed asphalt mixes. It is described in a research report, "Development of a Mix Design Process for Cold In-place Rehabilitation using Foamed Asphalt." They use a small laboratory asphalt-foaming plant (see Figure 13.25) to produce the foamed asphalt. This equipment simulates the foamed asphalt to be used during construction. The expansion and half-life characteristics of the foamed asphalt binder are determined for at least three asphalt binder temperatures and a range of water contents of 1–5 percent by weight of asphalt binder. The combination of the asphalt binder temperature and water content that maximizes the expansion ratio for the longest half-life is selected for use.

The optimum moisture content for the reclaimed asphalt pavement/base material is determined using standard procedures (modified Proctor) for determining the optimum density and moisture content for an unstabilized material.

To determine the optimum asphalt-binder content, samples of the reclaimed material are oven dried and cooled to room temperature and then rewetted to 90 percent of the optimum moisture content. A range of foamed asphalt contents is added to the reclaimed material, mixed thoroughly at room temperature, and compacted using either 50 or 75 blows with a Marshall hammer. After compaction, the samples are extruded from the molds. After extrusion, they are further cured for 3 days in an oven at 100°F (40°C). The density of the specimens is determined by measuring the bulk specific gravity. The samples should be handled very carefully after compaction. Because of the fragility of the samples, the bulk specific gravity should be determined using the CoreLok procedure discussed in Chapter 4 "Hot Mix Asphalt (HMA) Design." As described in Chapter 4, the samples are tested to determine the Marshall stability and the indirect tensile strength of the samples.

Additional companion samples are manufactured at each of the asphalt contents discussed above and soaked at 77°F (25°C) for 24 hours. They are surface dried and tested for strength using the same test procedure as that used on the dry samples. The water-soaked strength is compared

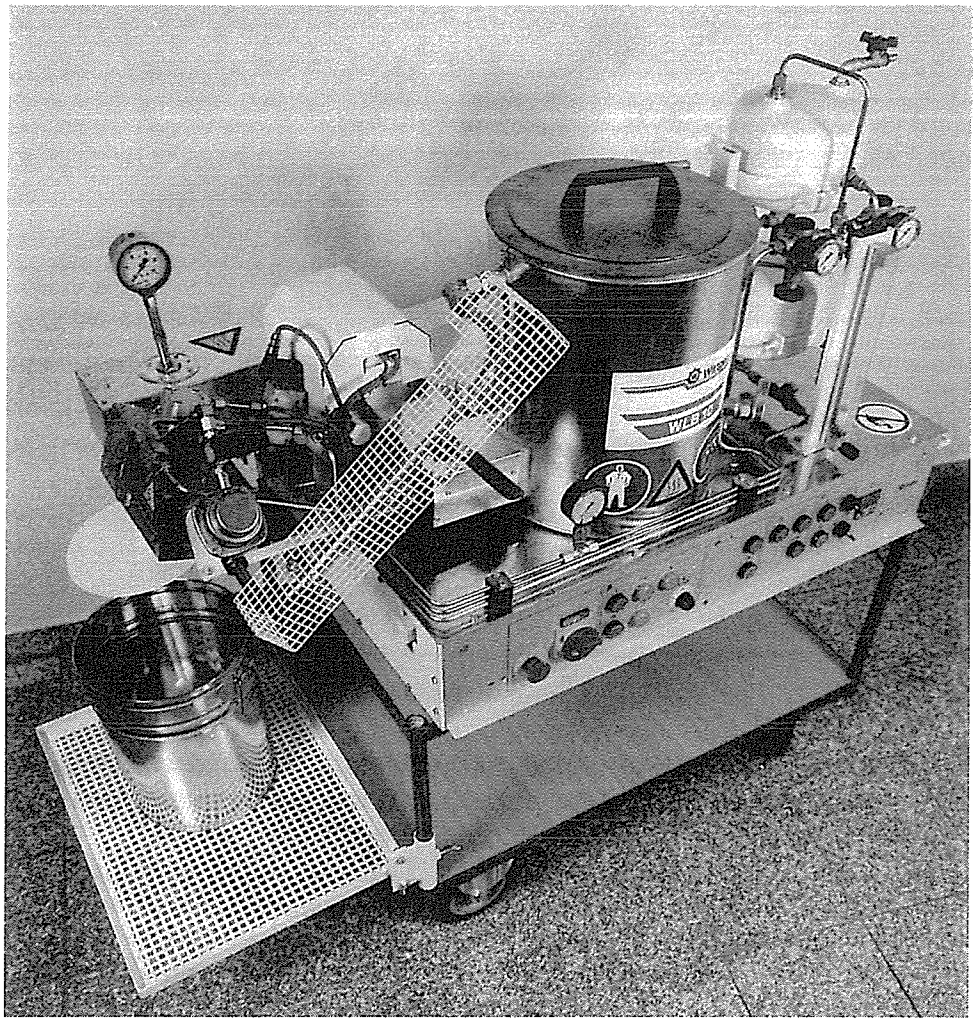


Figure 13.25 Laboratory Asphalt-Foaming Plant

with the dry strength and calculated to determine the moisture sensitivity of the mixture. *Optimum binder content* is defined as the binder content at which the soaked strength is the highest.

13.7.3 Construction of a FDR Pavement

Pulverization of the Existing Pavement

Before the introduction of modern reclaimers, the FDR of a pavement was done by scarifying the existing pavement system with a motor grader and large rippers attached to crawler tractors. This scarification was followed by the use of rotary mixers that pulverized the ripped material. The introduction of cold-milling machines led to the production of large, self-propelled high-powered reclaimers such as the one shown in Figure 13.26. Most reclaimers have an



Figure 13.26 Modern FDR Reclaimer

8 feet (2.4 meters) wide drum and can pulverize a maximum of 12 inches (300 millimeter) deep. During the pulverization process, the cutting drum will "up-cut," or rotate opposite to the direction the reclaimer travels.

In addition to the FDR reclaimer, the FDR process requires the use of a motor grader, to spread and fine-grade the reclaimed material, and compaction equipment.

Multiple passes of the reclaimer over the pavement section are necessary to ensure that the target depth is adequately pulverized and to pulverize the entire roadway width. This results in a series of longitudinal joints and some overlap between the passes. The minimum overlap width is normally 4 inches (100 millimeter).

The gradation of the reclaimed material should be monitored to ensure that the gradation being produced matches the mix design. The gradation of the reclaimed material is influenced by these factors:

- Front- and rear-door opening of the pulverization/mixing chamber. The more the door is closed, the longer the pulverized material is retained in the pulverization/chamber, which allows the larger chunks to be broken down.
- Position of the breaker bar setting in the pulverization chamber. The closer the breaker bar is to the cutting drum, the finer the gradation will be.
- The speed of the cutting drum. For thick lifts, a lower cutting drum speed is used, and for thinner lifts, a faster-cutting drum speed is used.
- The forward speed of the reclaimer. The slower the forward speed, the finer the gradation of the material will be.

- The condition of the existing pavement. When the existing pavement is badly cracked, sizing of the material will be more difficult.

The physical properties of the asphalt pavement are affected by the ambient temperature. If the pavement is cool, the asphalt layer is stiff and brittle and can be pulverized into small pieces very easily (see Figure 13.27). Increasing temperatures cause the pavement to become more plastic, and slabbing (asphalt lifting or breaking off in large pieces) can occur. The most efficient temperature range for sizing the reclaimed material is 50°–90°F (10°–30°C).

If additional pulverization or remixing is required for uniformity, the reclaimed material should be lightly compacted and reshaped. This process accurately controls the mixing depth. The rear wheels of the reclaimer compact or reduce the thickness of the fluffed-up material during pulverization. With light compaction and reshaping, the reclaimed material will have uniform thickness, which will allow better control of the treatment depth.

When an asphalt emulsion is added to the reclaimed material, a nurse truck is connected to the reclaimer (see Figure 13.28). The reclaimer is equipped with a meter capable of controlling the rate of flow and the total amount of asphalt emulsion being added to the reclaimed material. The meter should be linked to the forward speed of the reclaimer so that the amount of asphalt emulsion added will change according to the operational speed of the reclaimer. If asphalt emulsion is used, some agencies add lime or cement to improve the early strength of the mix and to speed up the curing process.



Figure 13.27 Pulverized Material Exiting the Reclaimer

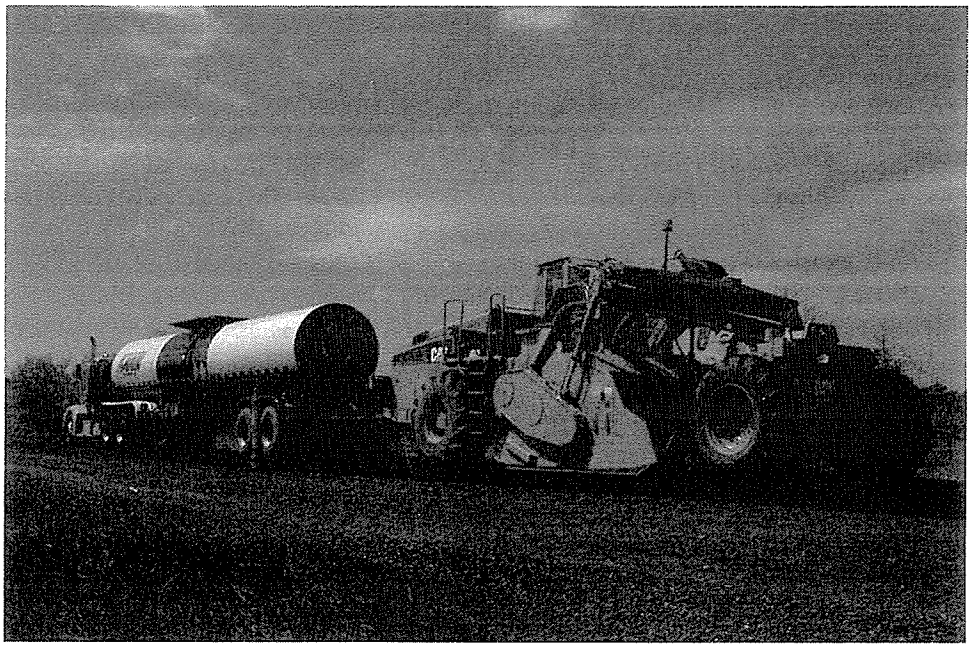


Figure 13.28 Reclaimer with Nurse Truck

If foamed asphalt is used, the reclaimer must be attached to a nurse truck and be equipped with a computerized onboard foam-generating system. It must also have a storage tank for the water used in the foaming process, and a system for injecting the foamed asphalt into the reclaimed material. The system should be linked to the forward speed of the reclaimer and the volume of the reclaimed material being processed.

13.7.4 Shaping and Compaction Procedures

A motor grader is used to move and place the reclaimed material to the desired longitudinal grade and cross-slope (see Figure 13.29). The amount of motor grader work required to place the reclaimed material depends on the original shape of the roadway, the specified final shape of the roadway, and the type of wearing surface that will be used. The final grade should be similar to that required of an untreated base course. The smoothness of the final HMA surface is directly related to the smoothness of the pulverized and reclaimed layer. To ensure a smooth final surface, tight blading or trimming the reclaimed surface may be necessary.

The reclaimed material is generally not at the optimum moisture content for compaction. Either aeration will be required to dry the reclaimed material, or moisture will have to be added to aid compaction. A water truck can be used as a supply truck to the reclaimer's onboard liquid additive, or it can be used to add water to the reclaimed material through a spray bar.

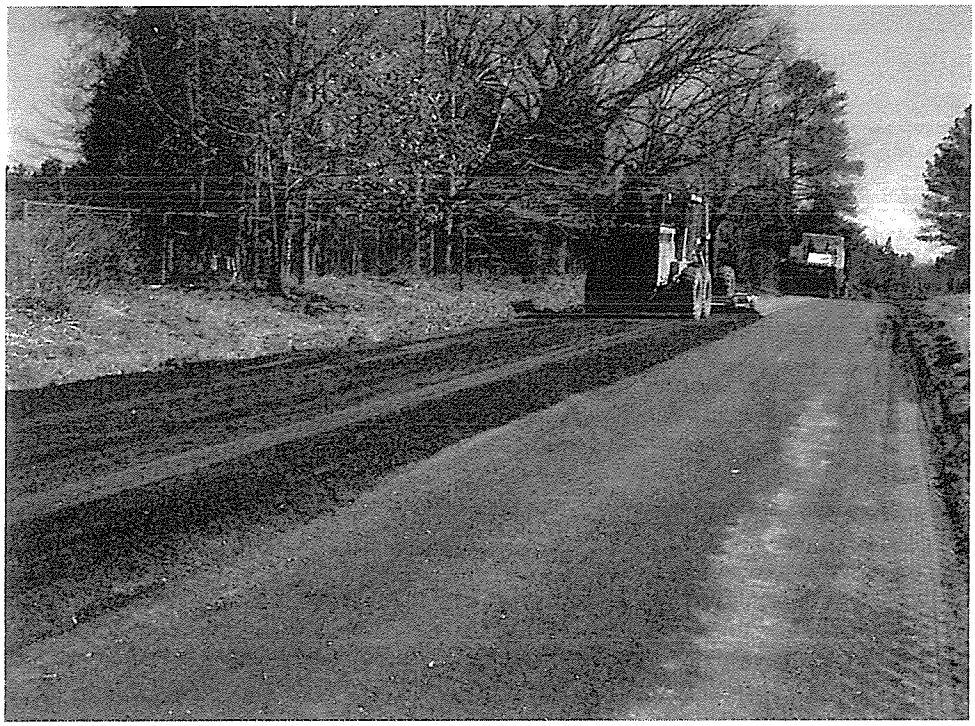


Figure 13.29 Motor Grader Shaping Reclaimed Surface

The level of compaction achieved in the reclaimed material is directly related to the performance of the final pavement. Figure 13.30 shows a reclaimed surface before compaction. Compaction of reclaimed mixes using an asphalt emulsion as a stabilizing agent must be accomplished soon after the emulsion begins to break. This is when the emulsion begins to change from brown to black, which also indicates that the moisture is evaporating from the reclaimed material. The compaction process becomes more difficult as the emulsion breaks.

Compaction of reclaimed mixes using foamed asphalt should occur shortly after placement of the mix. The mixes will remain workable as long as the moisture content stays at or above the moisture content at the time of foamed-asphalt injection and mixing. The compaction process should be completed before the foamed asphalt mix begins to dry out.

Compaction should be done with heavy pneumatic or steel-wheel rollers. The depth of the reclaimed material that must be compacted and the desired degree of compaction will influence the weight and the amplitude/frequency of vibration for vibratory compactors and the static weight and tire pressure of pneumatic rollers. The roller pattern should be established to determine the best mixture of roller types and number of passes.

As with CIPR mixes, FDR mixes using emulsions must be cured before a wearing course can be placed. The time required for this process depends on daytime and nighttime temperatures, the humidity levels and rainfall

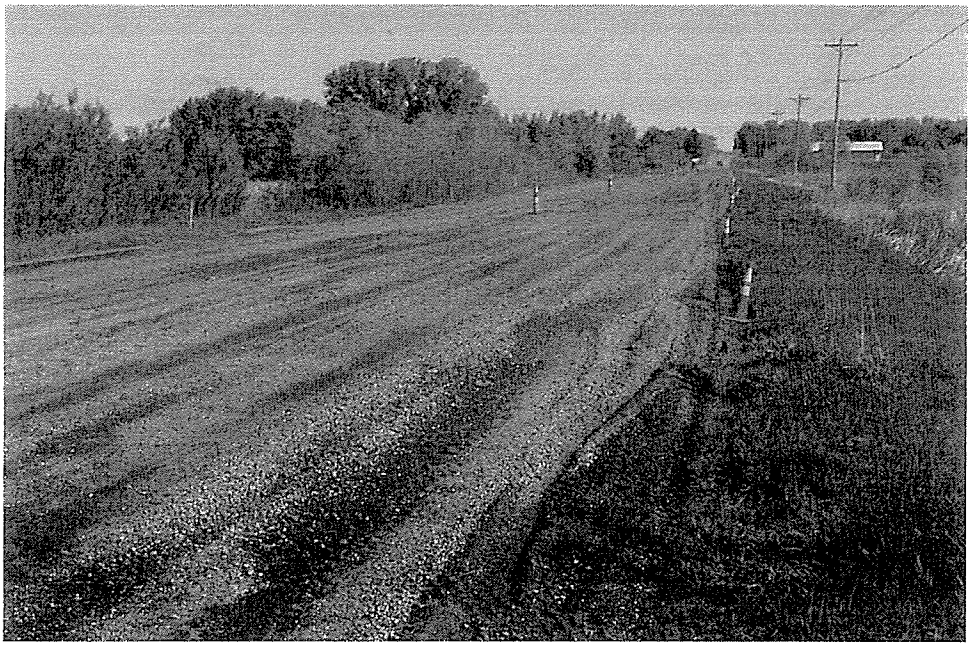


Figure 13.30 Reclaimed Surface Before Compaction

after completion of the recycling process, and the amount of water added to the mix during the FDR process. The goal is to ensure that the moisture content of the FDR mixture is reduced to about 2 percent before placing the wearing course.

13.7.5 Inspection

Prior to the start of any FDR project, the centerline of the existing roadway should be referenced by offset markers along the edge of the roadway outside the limits of construction. Such markers will facilitate the construction of the new pavement without the need for a survey layout crew.

During the construction process, the following should be checked:

- The treatment depth should be physically checked by removing the reclaimed material on either side of the reclaimer.
- The gradation of the reclaimed material should be checked periodically. This generally consists of checking the maximum particle size of the reclaimed layer. The sample should be taken from the full depth to ensure that the reclaimer is treating the entire depth of the layer being reclaimed.
- The yield rate of the emulsion or the foamed asphalt and additional aggregate should be computed and compared with the mix design.
- The FDR mix should be checked to ensure that it is homogeneous, with no segregation, and that the aggregate is coated with emulsion or foamed asphalt throughout its depth. It should also be checked to

ensure that no large (sometimes agglomerated particles or uncrushed pieces) are in the mix. If they are present, the pulverization process should be checked.

- Compaction should be checked using a nuclear density gauge. It should be 97 percent or greater than the control-strip density.
- The surface of the reclaimed mix should be checked for smoothness with a 16 feet (5 meters) straightedge with no area deviating more than ½–1 inch (13–25 millimeter) over the length of the 16 feet (5 meters) straightedge.

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CHAPTER 14

REHABILITATION OF CONCRETE PAVEMENTS WITH HMA

This chapter discusses techniques used to manage or eliminate reflection cracking in hot mix asphalt overlays of Portland cement concrete pavements.

Section 14.1 Introduction
14.1.1 Reflection Cracking

Section 14.2 Saw-and-Seal
14.2.1 Design Considerations
14.2.2 Construction

Section 14.3 Fractured Slab Techniques
14.3.1 Crack/Break and Seat
14.3.2 Construction

Section 14.4 Rubblization
14.4.1 Project Selection and Overview
14.4.2 Design Considerations
14.4.3 Construction
14.4.4 Specification Criteria and Inspection



14.1 Introduction

Many miles of Portland cement concrete (PCC) pavement are rehabilitated each year by placing an asphalt overlay on them. According to the FHWA's "Highway Statistics 2005," there were more than 40,000 more miles of composite pavements than rigid (PCC surface) pavements in the United States. Chapter 6, "Thickness Design" discusses the thickness design procedures for those overlays. This chapter discusses techniques used to either manage the reflection cracking in those overlays or eliminate the occurrence of reflection cracking.

14.1.1 Reflection Cracking

Reflection cracking is the result of horizontal and vertical movements at the joints and cracks in the underlying pavement, resulting in a pattern of cracking in the surface that reflects the joints or cracks in the underlying PCC pavement. Temperature changes in the pavement cause the slabs to contract and expand. This movement is most pronounced at joints and cracks, which open and close in response to the volume changes in the PCC slabs. Opening of joints and cracks due to cooling causes tensile stress directly over these locations in the HMA overlay (see Figure 14.1). Reflection cracking may occur after an extreme cold temperature event or after repeated opening and closing.

Traffic loading may produce differential vertical movement across the joint or crack, particularly when load transfer is inefficient. This causes shear and bending stress in the overlay. As the wheel (see Figure 14.2) approaches, the shear stress will reach a maximum at point A, and as it passes over the joint, it will reverse at point C. As the wheel load passes over the joint at point B, the bending stress will reach a maximum.

The key to delaying or avoiding reflection cracking is to reduce or eliminate the stresses and deformations produced in the overlay. Some treatments include the placement of an interlayer to reduce or dampen the movement at the joint or crack caused by thermal contraction of the PCC. Some of the techniques are discussed in Chapter 13, "Pavement Rehabilitation Methods for HMA Pavements," including the use of geotextiles, asphalt-rubber "stress-absorbing membrane interlayers" (SAMIs), etc. Single- and multiple-surface treatments have also been used as an interlayer for rehabilitation of HMA- and PCC-surfaced pavements, as have specialty asphalt paving mixtures that are extremely ductile and resistant to cracking. Another widely used approach is to control reflection cracking by sawing and sealing the overlay directly over the joints. This will be discussed in more detail later in this chapter.

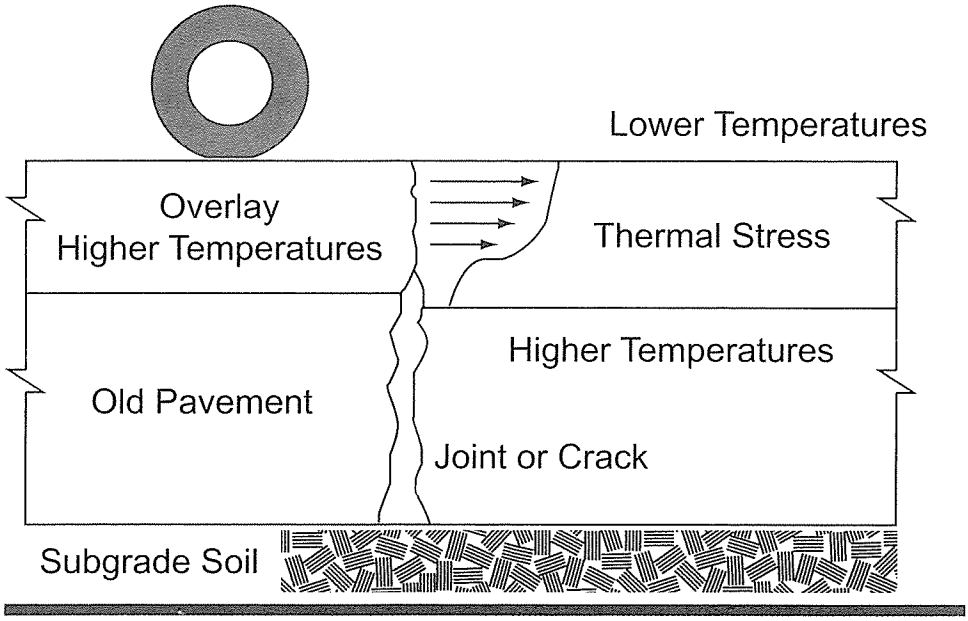


Figure 14.1 Stresses in an HMA Overlay Due to Temperature Changes

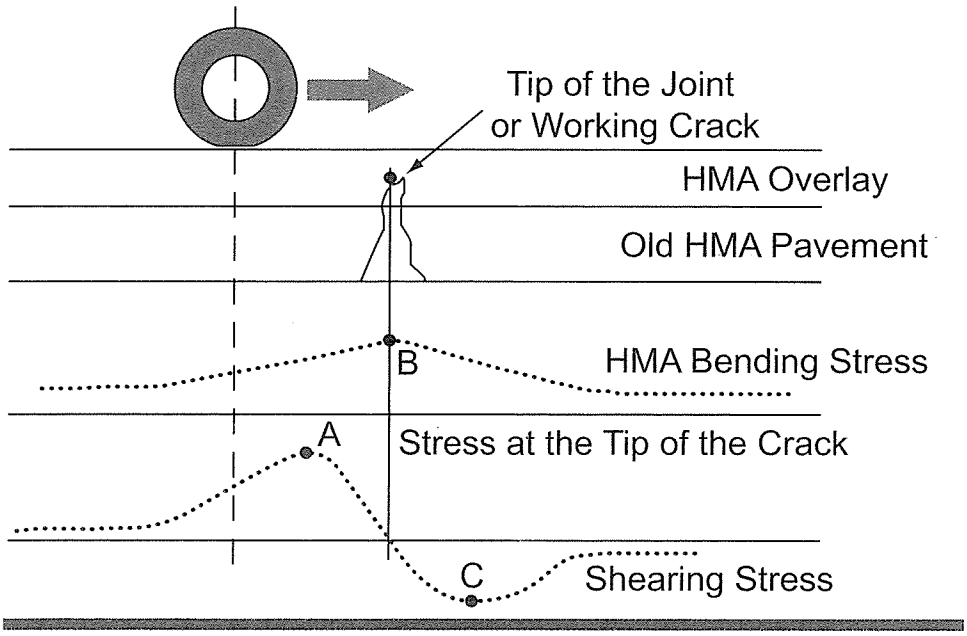


Figure 14.2 Shear Forces in Overlay Due to Traffic

These applications can work successfully when the existing pavement is in good structural condition (good load transfer, intact slabs). While it is not unusual to see a single lift of HMA used, two or more lifts are recommended where feasible to provide sufficient cover over the interlayer and to allow for the smoothest finished surface.

While an interlayer or saw-and-seal can be used effectively to reduce, delay, or control reflection cracking resulting from thermal movements, these treatments are not recommended where the load transfer across joints or cracks is poor. Fractured slab techniques that either reduce or eliminate slab action, such as cracking and seating and rubblization, are commonly used to address reflection cracking in structural HMA overlays of Portland cement concrete pavement (PCCP).

14.2 Saw-and-Seal

Saw-and-seal has been used for many years to slow the rate of deterioration of reflection cracks in relatively thin overlays of jointed PCCP. The technique consists of carefully referencing all the joints in the existing pavement before overlay and then sawing and sealing contraction joints in the HMA overlay directly over the underlying joints. When properly done, a weakened plane is formed directly above the joint defining the ultimate cracking pattern in the overlay. The saw-and-seal procedure will do the following:

- Provide maintainable joints
- Reduce or eliminate the spalling associated with reflection cracking
- Provide for good ride quality
- Provide a neat, well-maintained appearance

Saw-and-seal has most commonly been used on jointed-reinforced concrete (JRPC) pavements on highway and street projects in New England and Louisiana, as well as on airfield pavements. JRPC pavements generally have longer joint spacing than jointed-plain concrete (JCPC) pavements, resulting in longer slabs and thus greater opening and closing movement at the joints. It should only be considered for jointed concrete pavements that are in good structural condition, have good joint load transfer, and have intact panels (no working cracks) and minimal spalling. Thought should be given to how the joints can be practically referenced before selecting this approach, because missing the joint by more than one inch usually results in a jagged, irregular reflection crack alongside a neat sawed and sealed crack.

This technique should not be considered for the following:

- Overlays on JCPC that have a high frequency of mid-panel cracking (more than 20 percent) or badly deteriorated cracking. Panels with working cracks should either be patched or replaced with a new concrete panel.
- Overlays on JCPC with badly deteriorated joints and/or significant patching at or near the joints
- Overlays where there is insufficient load transfer across the joints as evidenced by ¼-inch (0.0625 millimeters) faulting and/or pumping. When tested using nondestructive testing equipment, joints should have a minimum 60 percent load-transfer efficiency.
- Overlays of pavements with wide joints—greater than ¾ inches (19 millimeters)

14.2.1 Design Considerations

Minimum Thickness

Saw-and-seal is commonly considered when the overlay thickness that can be placed is limited. Examples include urban streets with curb and gutter sections, intersections and driveways, and highways with vertical clearance restrictions. As previously suggested, an overlay is best applied in two lifts to provide the greatest opportunity to improve ride quality and surface drainage, but there have been numerous examples where a single lift of HMA has been placed, has been sawed and sealed, and has provided excellent performance. The minimum recommended thickness for a single lift of HMA is 2 inches (50 millimeters). For two lifts, the minimum thickness is 3½ inches (90 millimeters).

Joint Configuration

Many state DOTs have specifications for this process. The engineer should check with the local DOT to ensure that his or her specification follows their requirements. The typical requirement for the joint is that the HMA layer be cut to one-third to one-half of the thickness of the HMA layer. This can range from ⅛ inch wide (if a reservoir for the sealant is used) to ½ inch wide if no reservoir is used. Figure 14.3 presents the procedure most DOTs utilize, which is the ⅛ inch cut with a reservoir for the joint sealant material.

Urban and Other Areas

When using saw-and-seal in urban areas where curb and gutter sections are present, the saw cuts should line up with the control joints in the gutter sections, even if a slightly skewed saw cut results. The saw cuts should not be spaced more than 40 feet (12 meters) apart, and for thin JPCP, it may be beneficial to have the cuts spaced closer, such as 30 feet (9 meters) apart. For parking lots, airports, and other large areas of pavement, saw-and-seal has been shown to be effective when the saw cuts are made so that the pavement is cut into 50-foot (15-meter) squares.

14.2.2 Construction

Preparation Before Overlay

Before placing an HMA overlay, the existing concrete pavement should be repaired as needed. Concrete panels with working cracks, corner breaks, moderate-to-severe spalling, and joints exhibiting poor load transfer should be repaired before placing the overlay. Referencing the existing joints in the concrete pavement is critical. Fixed reference markers should be used. Chalk or paint is used to mark the joint (see Figure 14.4) after the HMA overlay is placed. A deviation of 1 inch (25 millimeters) can result

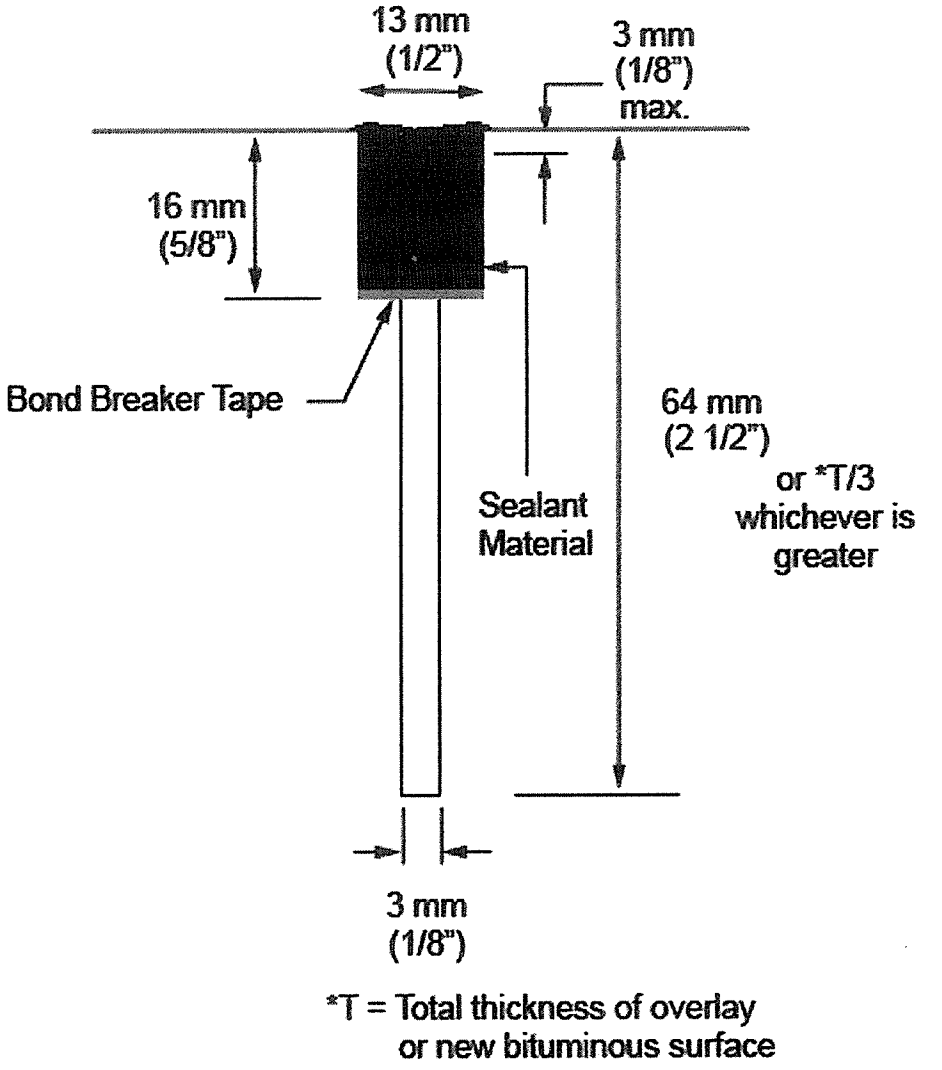


Figure 14.3 Deep Saw Cut with Reservoir

in an uncontrolled, jagged reflection crack forming adjacent to the sawed joint (see Figure 14.5).

Timing

The saw-cutting and sealing should be done as soon as possible after placement (the next morning) of the HMA layer because the saw can be operated without damaging the HMA layer, but it should be done no later than 48 hours after the wearing course is placed. The saw-cutting might possibly damage the joint edges, resulting in poor sealant adhesion and joint spalling. The concern is that the joint will reflect through before the saw cut is made. The saw-cutting can be done with either wet or dry cutting. Sawing and sealing should not be done if the pavement is wet or if the ambient temperature is less than 40°F (4°C).

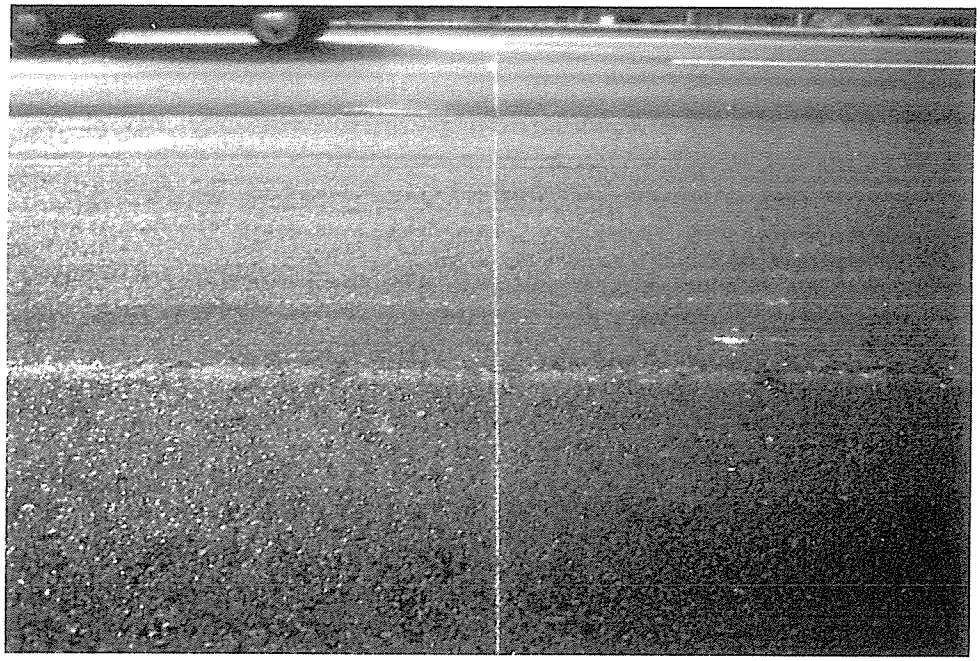


Figure 14.4 Marked Joints Prior to Sawing



Figure 14.5 Secondary Reflection Crack

Shoulders

It is recommended that the saw-cut extend at least 12 inches (300 millimeters) beyond the edge of the JCPC layer being overlaid into the shoulder, on pavements with HMA shoulders. On pavements with gravel shoulders, the saw-cut should stop just short of the outer edge of the HMA. This prevents the sealant from flowing out of the joints and onto the shoulder.

Sealing of Saw-Cut

The sawed joint must be clean and dry when the backer rod and sealant material are placed. The sealant material should meet the requirements of ASTM D 6690. If dry sawing is used, the joint should be cleaned with compressed air to remove the dirt, dust, and other deleterious material. If wet sawing is used, the cut should be flushed with high-pressure water until it runs clear. It then should be dried using compressed air.

The sealing of the saw-cut should follow these steps (see Figure 14.6):

- Install the backer rod at the specified depth. If the reservoir style is used, a backer rod may not be needed, but steps should be taken to ensure that the sealant does not enter the remainder of the joint.
- The hot sealant should be placed in the saw-cut according to the manufacturer's recommendations.
- The sealant material should fill the joint so that after cooling, the level of the sealing is 1/16–1/8 inches (1.6–3.2 millimeters) below the pavement surface.
- Traffic should be kept off the sealed joints until the sealant has cured. If traffic must be placed on freshly sealed joints, it may be necessary to cover the joints with fine sand or tissue paper.

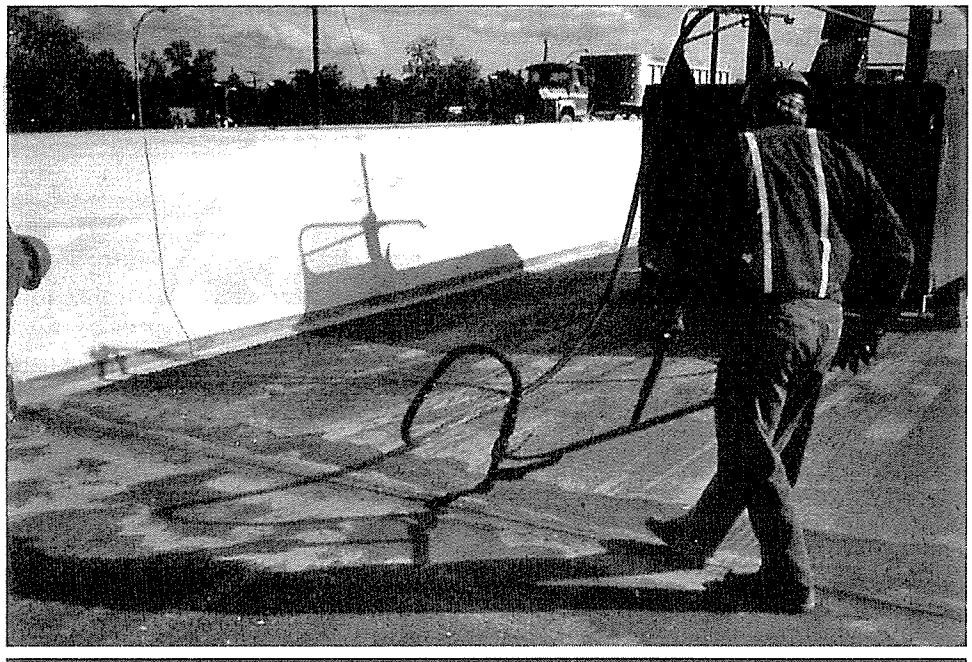


Figure 14.6 Placement of Sealant

14.3 Fractured Slab Techniques

Two procedures have been developed to fracture PCC slabs, thereby reducing the stresses and deformations that occur in an asphalt overlay—crack/break-and-seat and rubblization. The crack/break-and-seat procedure reduces the effective size of the PCC slabs and seats them firmly onto the underlying layers with heavy rollers. The rubblization process eliminates slab action by reducing the PCCP to aggregate-size pieces, resulting in a layer that functions in a similar way to an aggregate base course.

14.3.1 Crack/Break and Seat

Crack/break and seat minimizes joint and crack movement by shortening the effective slab length and firmly seating the broken PCC pieces onto the underlying layer (see Figure 14.7). The PCC is broken using a drop hammer, and then seated with a large, ballasted rubber-tire roller. The intent is to create slabs that are small enough to reduce the movement at the cracks in the PCC layer that were caused by low temperatures and daily temperature variation, while still being large enough to develop flexure under loading.

Cracking and seating is used on JCPC pavement, while breaking and seating is used when the PCC pavement is JPCP. The broken slabs are seated to re-establish firm support between the underlying layers and the slabs, reducing the amount of differential vertical movement under traffic.

In either operation, the slabs must be cracked through the entire slab in order to be effective; for breaking and seating, the reinforcing steel

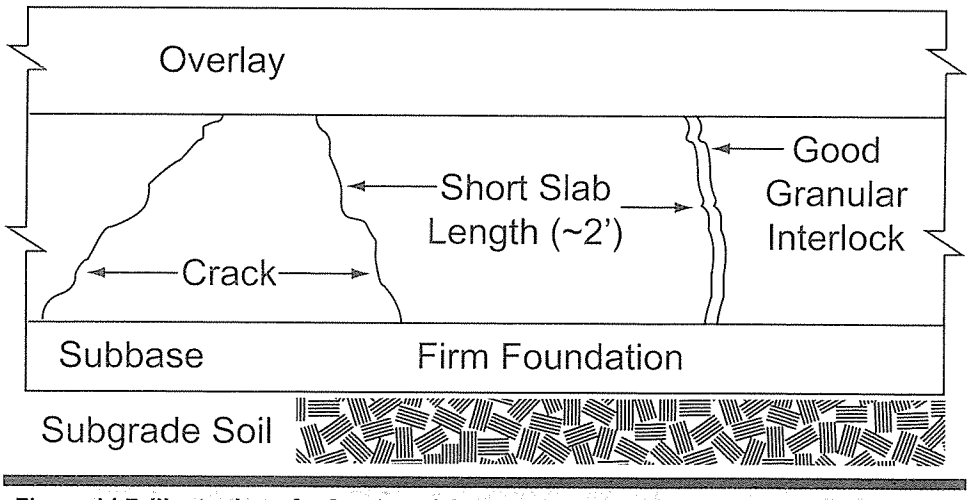


Figure 14.7 Illustration of a Crack-and-Seat Pavement

or wire mesh must be ruptured. Historically, cracking and seating has performed very well, but this has not been the case with breaking and seating because of the difficulty in ensuring that the steel has been ruptured. This problem has led many agencies to use rubblization instead of breaking and seating when rehabilitating joint reinforced concrete (JRC) pavements. Breaking and seating should only be considered for relatively thin (less than 8 inches; 200 millimeters) or variable thickness (often called "thickened-edge") JRC pavements.

Some agencies have used various types of interlayers or specialty mixtures in conjunction with cracking and seating. The interlayer provides added insurance against reflection cracking, and some can help seal the pavement structure. When a fabric or a SAMI is used, it is often recommended to place these directly on a leveling course of HMA rather than directly over the fractured PCC layer. If an open-graded crack relief layer is used, though, it should be placed directly on the fractured PCC, and provision must be made for removing any water collected within this layer by daylighting or using longitudinal drains.

Project Selection and Overview

Cracking/breaking and seating has been shown to be an effective technique when the following distresses occur in the PCCP pavement:

- Seriously faulted transverse joints and cracks
- Rocking of slabs due to the presence of voids
- Presence of working transverse and longitudinal cracking
- Uneven slab settlement
- Corner breaks
- Blowups, shattered slabs

The quality of the underlying layers should be thoroughly investigated during the design process. It has been determined that the performance of the crack-and-seat process is not very good when the PCC was placed directly on a fine-grain subgrade, particularly in wet conditions. It is important that a thorough sampling and testing program be conducted during the design process. The thickness of the concrete pavement must be determined by cores. The properties of the subgrade should be determined by cutting test pits in the existing PCC pavement to determine the type, thickness, and strength of the base and subgrade materials. These underlying materials should be evaluated in the laboratory to determine their strength, using R-Value, CBR, or elastic modulus testing. The base materials should be evaluated to determine their drainability if a subdrainage system is to be used. The subgrade materials should also be evaluated for potential swelling problems.

Design Considerations

The overlay should be designed according to the procedures outlined in Chapter 6. If the existing PCC pavement shows signs of pumping, installation of a subdrainage system to drain the water away from the pavement structure should be seriously considered. The Asphalt Institute recommends a minimum overlay thickness of 5 inches (125 millimeters) of HMA over cracked or broken and seated PCC.

14.3.2 Construction

The objective of the crack-and-seat process is to break the pavement into slabs ranging in size from 18 to 48 inches (0.5 to 1.5 meters). Breaking the pavement into the smaller sizes has shown to provide better performance. The PCC slab must be cracked full depth. A test strip should be established to ensure that the equipment and processes used are sufficient to thoroughly crack the PCC pavement. Cores should be drilled directly over cracks to verify that the slab is being cracked throughout the entire depth of the layer.

Cracking and Seating JCPC Pavement

Various types of equipment have been used to crack JCPC, including pile drivers, guillotine hammers, whip hammers, and impact hammers. Production rates will vary with the equipment type and the thickness of the pavement. The most common method used is the guillotine hammer (see Figure 14.8), which has high productivity and covers the full width of a lane.

After the JCPC has been cracked, the pieces must be firmly seated onto the underlying layer to prevent rocking under traffic. Seating is accomplished with a large pneumatic roller that forces the broken slabs into full contact with the foundation. Two passes of a 50-ton (450 kilonewton) roller are specified for the seating operation. Excessive rolling should be avoided because it can result in a loss of interlock between the cracked pieces or in shear failure of the existing subbase or subgrade.

Breaking and Seating JRCP Pavement

As previously defined, breaking and seating is conducted on existing JRCP pavement, and the process must rupture the reinforcing steel to be effective. This process requires that the slabs be broken into pieces smaller than those required for crack and seating. For breaking and seating, it is recommended that the slabs be broken into pieces measuring 12–18 inches (300–460 millimeters). The steel should be cut by transverse sawing of the pavement at routine intervals to ensure debonding in the transverse direction. The seating process is accomplished with a heavy pneumatic roller in a manner similar to that used in the crack-and-seat process, except that additional passes are needed to ensure rupturing of the steel. Five to seven passes of a 35–50 ton (315–450 kilonewton) rubber tire roller are recommended. Any soft spots or local failures identified in the seating process require careful removal and replacement. It is important to address drainage or unsuitable foundation conditions during this process. The broken PCC should be replaced with a full-depth HMA patch. The areas to be replaced should be sized to accommodate the equipment necessary to properly place and compact the new material.

Inspection

It is recommended that test sections be constructed to establish the cracking pattern. The goal is to achieve through-slab cracking and acceptable segmentation of the JCPC without excessive penetration, spalling, or shattering of the existing surface. If this occurs, the process should be



Figure 14.8 Guillotine Hammer Being Used for Crack and Seating JCPC Pavement

stopped and adjustments made in the cracking and/or rolling process. It is suggested that the contractor furnish and apply water for random checks of the crack pattern at least twice every day. The size of the broken slabs can be verified by spraying water on the surface of the cracked pavement (see Figure 14.9). The last part of the pavement to dry out will be the cracks, thereby revealing the shape of the crack pattern.

During the construction of the test section, it is recommended that some of the cracks be cored (6-inch diameter, 150 millimeters) to insure that the cracking is through the entire slab thickness.



Figure 14.9 Crack Pattern of Cracked and Seated Pavement